

Literature Benchmark Brief: Bremnes_2004_WindEnergy

Title: Probabilistic wind power forecasts using local quantile regression

Authors: P. K. Bremnes (2004)

Venue: Wind Energy | **DOI:** 10.1002/we.107

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Local linear quantile regression to estimate non-linear wind power quantiles

Empirical Scale & Experimental Dataset

Danish wind power dispatch data

Relevance to IEEE TSTE & Sustainable Energy Systems

First rigorous application of quantile regression to wind power uncertainty

Identified Gaps & Operational Vulnerabilities

Evaluates isolated single turbines without spatio-temporal array wake coupling

Role in Manuscript Defense & Reviewer #2 Proof

Cited for quantile loss formulation and pinball scoring criteria